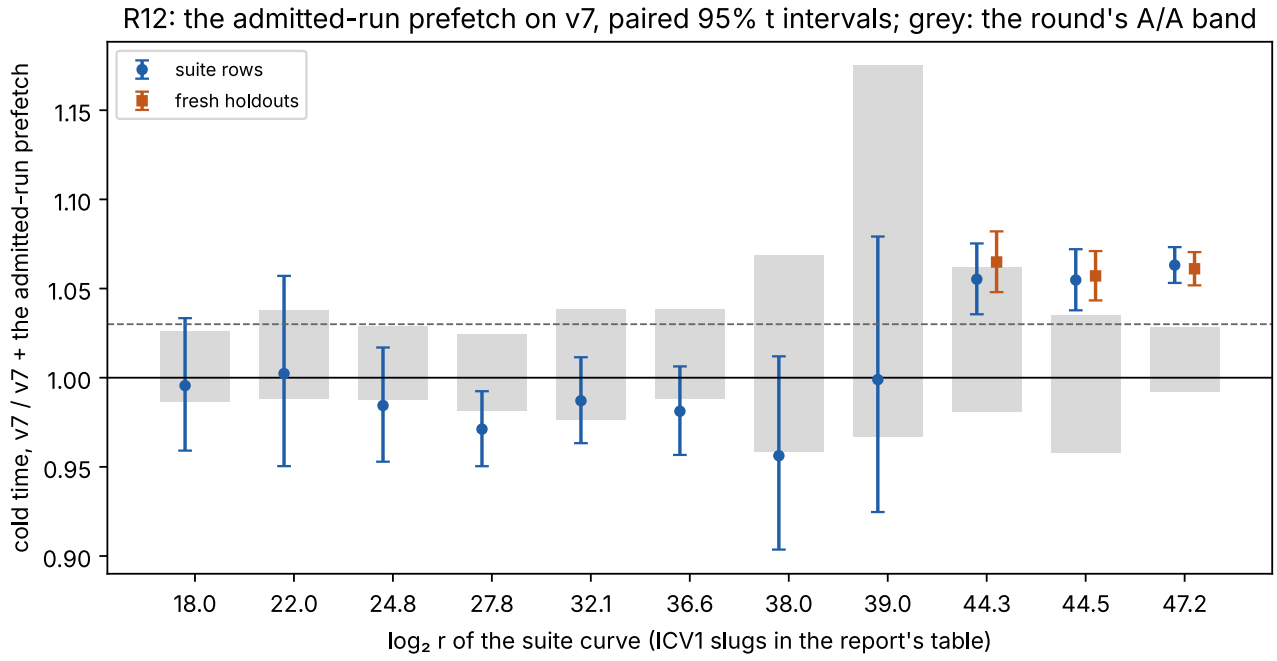
R12: the collection scan's admitted keys, their runs prefetched

Accepted, as baseline v8 (engineering). v7 (v6 plus R11's streamed build) against v7 plus R12's candidate (`candidate-v7.patch`: a second pass between the filter and the lookups prefetches each admitted key's run), both built with `rustc 1.97.0`, on an Emerald Rapids host of the reference class, 2026-10-16/17. Protocol: `research/ic_tool_program/rounds/R12-admitted-prefetch/PROTOCOL.md` (declaration, amendment 1); analysis: `analysis.json`; run trees: `runs.tar.xz` (SHA-256 `77d7953eaf604705…`).

- **Tests and pin:** 207 of 207 tests on the candidate's tree; every pinned output v0's on all 90 rows.
- **Timed:** 864 isolated processes (A/A 222, suite 402, holdouts 240), 4 contended and re-run, none failed; the driver task was relaunched once on the same container.
- **Decision:** at each target size the cold interval lies above 1.03 on the suite rows and the fresh holdouts; no size regresses beyond its A/A band. Predicted 1.06–1.11, 1.10–1.17 and 1.06–1.12: the round measured the low end, or below it at 2^44.5.
- **The smaller sizes cost a little.** From 2^24.8 to 2^38.0 the point estimates read 0.956–0.987, and at 2^27.8 the interval lies below 1 (0.971 [0.950, 0.992]). That is inside the declared A/A rule, but it is a real cost of the extra pass where the table is small. Gating the pass on the table's size is the next candidate.



research/ic_tool_program/rounds/R12-admitted-prefetch/analysis.json; 864 isolated processes, 4 contended (excluded); dashed: the 1.03 acceptance line

Figure 1: Cold time, v7 over the candidate (above 1, the candidate is faster), paired 95% t intervals; grey: the round's own A/A band; dashed: 1.03. Values from `analysis.json`.

Suite rows (five rounds)

curve	$\log_2$ r	rows	cold [95%]	collection	S cold, v7 → v8	A/A band
`icv1-f2m19-tm797-9c54981b`	18.002	2	0.996 [0.959, 1.033]	1.033	119.29 → 121.02	0.986–1.026
`icv1-f2m23-tm5197-1f85e9e1`	22.001	2	1.002 [0.95, 1.057]	1.008	31.39 → 30.68	0.988–1.038
`icv1-f2m45-tm6236725-40939294`	24.803	2	0.984 [0.953, 1.017]	0.971	18.5 → 19.28	0.988–1.029
`icv1-f2m37-tm534059-32aad96b`	27.781	2	0.971 [0.95, 0.992]	0.976	8.31 → 8.41	0.981–1.024
`icv1-f2m43-tm998717-e2e742b0`	32.113	2	0.987 [0.963, 1.011]	0.978	4.36 → 4.43	0.977–1.038
`icv1-f2m47-t22705043-f4e44623`	36.636	2	0.981 [0.957, 1.006]	0.961	2.34 → 2.36	0.988–1.038
`icv1-f2m57-tm747311035-c1f545af`	38.002	2	0.956 [0.904, 1.012]	0.937	2.5 → 2.54	0.958–1.068
`icv1-f2m41-tm2308219-7f48b14a`	39.0	2	0.999 [0.925, 1.079]	1.005	3.16 → 3.11	0.967–1.175
`icv1-f2m53-tm56619371-dac20a85`	44.259	8	1.055 [1.036, 1.075]	1.107	3.35 → 3.19	0.981–1.062
`icv1-f2m59-tm943548413-98844ecc`	44.517	8	1.055 [1.038, 1.072]	1.093	3.01 → 2.87	0.958–1.035
`icv1-f2m61-t158598901-ab42b6c5`	47.211	8	1.063 [1.053, 1.073]	1.078	4.42 → 4.17	0.992–1.028

Fresh holdouts (seeds 238–241, T167–T174)

curve	$\log_2$ r	pairs	cold [95%]	S cold, v7 → v8
`icv1-f2m53-tm56619371-dac20a85`	44.259	40	1.065 [1.048, 1.082]	3.29 → 3.12
`icv1-f2m59-tm943548413-98844ecc`	44.517	40	1.057 [1.043, 1.071]	3.04 → 2.89
`icv1-f2m61-t158598901-ab42b6c5`	47.211	40	1.061 [1.052, 1.07]	4.42 → 4.13

Scope

One host and one core, the reference class. No value the scan computes changes, so the counts and the ratio to the floor are unchanged. Chained with R06, R08 and R11, cold time is 1.90–1.98× faster than v4 at the three target sizes (a product of paired ratios). The distance to matched rho is the rule at v7′s until a rule runs at v8.